A MORPHISM FROM AN AFFINE SPACE OVER THE BASE TO A SPACE WITH AFFINE DIAGONAL IS AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof labels its steps 'Proof of (1)', 'Part (2) follows from (1)' and so on; all four assertions are handled inside the single counted proof environment.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Affineness is stable under base change. Lemma lemma-base-change-affine, used at proof lines 4133-4138. Makes both X x_B Y -> Y and the graph (1, g) affine.
- A composite of affine morphisms is affine. Lemma lemma-composition-affine, used at proof lines 4139-4140. Concludes affineness of g from affineness of its two factors.
- A closed immersion is affine. Lemma lemma-closed-immersion-affine, used at proof lines 4141-4142. Turns the separatedness hypothesis into the affine diagonal hypothesis.
- Separatedness of an algebraic space over B means the diagonal is a closed immersion. Properties of Spaces, Definition spaces-properties-definition-separated, used at proof lines 4124-4126. Identifies the two hypotheses in parts (2) and (4).

Source reference. Original lemma 03WI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5113–5113. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 07U2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9957–9957. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03WH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5112–5112. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original definition 03BS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4359–4359. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let B be an algebraic space over S. Suppose $g : X \to Y$ is a morphism of algebraic spaces over B.*

(1) *If X is affine over B and $\Delta : Y \to Y \times_B Y$ is affine, then g is affine.*
(2) *If X is affine over B and Y is separated over B, then g is affine.*
(3) *A morphism from an affine scheme to an algebraic space with affine diagonal over $\mathbf{Z}$ (as in Properties of Spaces, Definition 03BS) is affine.*
(4) *A morphism from an affine scheme to a separated algebraic space is affine.*

Proof. Proof of (1). The base change $X \times_B Y \to Y$ is affine by Lemma 03WI. The morphism $(1, g) : X \to X \times_B Y$ is the base change of $Y \to Y \times_B Y$ by the morphism $X \times_B Y \to Y \times_B Y$. Hence it is affine by Lemma 03WI. The composition of affine morphisms is affine (see Lemma 03WH) and (1) follows. Part (2) follows from (1) as a closed immersion is affine (see Lemma 07U2) and Y/B separated means Δ is a closed immersion. Parts (3) and (4) are special cases of (1) and (2). □